

Water Management System

Getting started

General

The Water Management System (WMS) is an online platform that monitors and manages connected AQtap water dispensers. WMS offers access to:

- data on water consumption
- operational status and service logs
- reports on water credit transactions
- setting of water prices
- distribution of water credits.

WMS safety

Each owner of one or more AQtaps receives a proprietary WMS group with a unique ID. Each user has access to his group only.

Self-registration in GRUNDFOS LOGIN

New owners of AQtaps receive a welcome email with a link (URL) to WMS. To register as new user:

1. Click on the link to WMS.
 - > The GRUNDFOS LOGIN page opens.
2. Click on "Not registered yet?" to create a new login.
3. Enter the requested data:
 - email address as login name (same as for order),
 - password,
 - name.
4. Read and confirm "Privacy policy" and click on "Register".
 - > A confirmation email for user profile validation is sent to the login email address.
5. Validate user profile by clicking on the link in the confirmation email.
 - > The login for the WMS group is created.

Accessing the WMS

1. Click on the WMS link in the welcome email.
2. Enter the login credentials:
 - Email-address,
 - Password.

WMS is now opened.

Operating of offline AQtaps

An AQtap water dispenser is fully operational, even if the connection to WMS cannot be established. If not connected to WMS, AQtaps cannot exchange data. All settings must be made directly at the AQtap water dispenser.

Setting up AQtap for online use

For exchange of data, the connection between AQtap and WMS must be established. The 2G GPRS communication must be set up, and the AQtap must be registered by the owner.

Setting up the GPRS communication

1. Conclude a mobile subscription with an approved mobile phone operator (telco).
 - Mini-SIM card (15 x 25 mm)
 - 2G machine-to-machine GPRS data subscription
 - Ability to send and receive international SMS messages
 - Ability to acquire a signal from the network operator you expect it to use
 - GPRS data subscription of 20 MB per month with automatic top-up
2. Disable PIN protection of the SIM card by setting it to "0000", e. g. using a cell phone. PIN protected SIM cards cannot connect to WMS.
3. Insert the SIM card in the power-off AQtap. The SIM card will only be read after power on.
4. Commission the AQtap using the ServiceCard.
5. Make sure that the AQtap is connected to GSM network.
 - GSM LED on CIM (upper LED, close to SIM card slot) is flashing yellow with 3 sec interval.

Commissioning is described in the installation and operating instructions provided with the AQtap.

Registering an AQtap in WMS

1. Make sure that the AQtap is connected to GSM network. This can reduce the time of the confirmation process.
2. Go to Settings > Dispensers > Add new dispenser
3. Set water price: enter amount of water credits per litre of water.
4. Register SIM card: enter phone number with country code. Enter IMEI number from HMI unit inside AQtap cabinet door. Select telco operator.
5. When the AQtap is commissioned, it confirms the connection to WMS.
 - > The confirmation process can take up to several hours, depending on the GSM network status on AQtap installation site.
 - > The connection status of a registered AQtap can be found under Report > Status map.

AQtap

Getting started

AQtap safety concept

AQtap water dispensers are provided with a number of safety measures to avoid misuse by invalid smart cards:

- Pre-activated AdminCards and ServiceCards will only be part of a Starter Package from Grundfos.
- All further smart cards must be ordered separately and activated at the AQtap.
- Smart cards can be used at all AQtaps within their group but not within another group.
- AdminCards and ServiceCards are code-protected. The code must be set at the AQtap when the cards are used for the first time. The set code cannot be changed afterwards.
- All smart cards can be blocked permanently via WMS.

Commissioning of an AQtap

AQtap water dispensers can only dispense water after commissioning. The AQtap is automatically assigned to the same WMS group as the ServiceCard used for commissioning. Commissioning is described in the installation and operating instructions.

Setting and changing the water price

The water price can be set in WMS or in the service menu of the AQtap. Setting the water price is described in the installation and operating instructions.

Activating new smart cards

All actions at the AQtap require activated smart cards. A new smart card must be activated with an activated smart card:

1. Place the activated smart card on the right card reader of the commissioned AQtap.
For an AdminCard enter the code.
 2. Place the new smart card on the left card reader.
 3. Press .
 4. Press .
- > Four flashes show that the new smart card is activated.

Downloading water credits at the AQtap

Water credits purchased via mobile payment or distributed from WMS must be downloaded to the assigned smart card.

1. Place smart card on left card reader.
2. Press .
 - > connection attempt is displayed
 - > number of available water credits is displayed.
3. Press  to start downloading the water credits.
 - > Four flashes confirm successful download.
 - > New water credit balance is displayed.

Transferring water credits between smart cards

Water credits can be transferred from VendorCards or WaterCards to WaterCards, and from AdminCards to VendorCards. Transferring water credits is described in the installation and operating instructions.

Smart cards and user profiles



The **water service provider** holding the **AdminCard** can:

- activate new VendorCards and ServiceCards
- create water credits on activated VendorCards



The **water vendor** holding the **VendorCard** can:

- activate new WaterCards
- download water credits from WMS
- transfer water credits to a WaterCard
- draw water from the AQtap for water credits



The **service technician** holding the **ServiceCard** can:

- run commissioning and set water price
- conduct a site acceptance test
- reset the AQtap



The **water consumer** holding the **WaterCard** can:

- draw water from the AQtap for water credits
- download water credits from WMS
- transfer water credits between WaterCards